Qi Le

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Efficient Machine Learning, Distributed Machine Learning
To develop cutting-edge AI products and services

Education

- **University of Minnesota, Twin Cities** | *Ph.D. Computer Science* — Sep 2022 – Present
- **University of Minnesota, Twin Cities** | *M.S. Data Science* — Aug 2020 – May 2022
- **Rensselaer Polytechnic Institute** | *B.S. Chemical Engineering, Minor in CS* — Aug 2016 – May 2020

Research Experience

- **Research Assistant**
  University of Minnesota, Twin Cities — May 2021 – Present
  Advised by Prof. Ali Anwar and Prof. Jie Ding.

  Efficient and Trustworthy Foundation Models

  - Designed **Probe Pruning** for online, dynamic, and structured LLM pruning using lightweight model probes and historical activations, without fine-tuning; released open-source code.
  - Provided design input for **GISP**, which iteratively prunes attention heads and MLP channels globally using task-aligned importance scores.
  - Collaborated on the design of **RESP**, which uses self-generated reasoning traces and progressive trace regeneration to preserve reasoning during structured pruning.
  - Proposed **OBMerge**, an iterative framework that merges task-specific models through parameter replacement guided by estimated loss changes, reducing parameter conflicts without joint training.
  - Introduced **ProjCache** to compress KV caches with output-projected eviction scoring and compensation inspired by Optimal Brain Surgeon, reducing attention-output reconstruction error.
  - Contributed to the method design of **MAP**, a novel, first-principles approach to aligning language models with multiple human values in a structured and reliable way.

  Federated and Collaborative Learning

  - Built **DynamicFL**, a communication-adaptive framework for addressing statistical heterogeneity in federated learning.
  - Developed **PFR**, a personalized federated recommender framework with private and partially federated autoencoders.

  Self-Evolving Agents and Skill Distillation *Ongoing*

  - Improving AI agents through evolutionary search that iteratively edits tools and execution harnesses to enable recursive self-improvement.
  - Exploring the distillation of agent harness execution traces and reusable skills into model parameters.

Publications

- **Q. Le**, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar, **OBMerge: Optimal Brain Merge via Iterative Parameter Replacement**, *Manuscript under review, 2026.*
- **Q. Le**, E. Diao, Z. Wang, X. Wang, L. Yang, J. Ding, A. Anwar, **ProjCache: Output-Projected KV Cache Eviction with OBS-Inspired Compensation**, *Manuscript under review, 2026.*
- Z. Wang, E. Diao, **Q. Le**, et al., **Think Before You Prune: Self-Reflective Structured Pruning for Reasoning Language Models**, *DAC 2026*
- Z. Wang, E. Diao, **Q. Le**, et al., **From Local to Global: Revisiting Structured Pruning Paradigms for Large Language Models**, *ACL 2026* Outstanding Paper

- J. Tang, X. Wang, **Q. Le**, et al., **AHE: Adaptive Homomorphic Encryption for Customizable Privacy in Heterogeneous Federated Learning**, *IEEE INFOCOM 2026*
- X. Wang, J. Du, A. A. Khan, **Q. Le**, et al., **Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models**, *NeurIPS 2025* Spotlight
- **Q. Le**, E. Diao, Z. Wang, et al., **Probe Pruning: Accelerating LLMs through Dynamic Pruning via Model-Probing**, *ICLR 2025*
- X. Wang, **Q. Le**, A. Ahmed, et al., **MAP: Multi-Human-Value Alignment Palette**, *ICLR 2025* Oral
- X. Wang, E. Diao, **Q. Le**, et al., **AID: Adaptive Integration of Detectors for Safe AI with Language Models**, *NAACL 2025*
- A. F. Khan, X. Wang, **Q. Le**, et al., **IP-FL: Incentive-Driven Personalization in Federated Learning**, *IEEE IPDPS 2025*
- **Q. Le**, E. Diao, X. Wang, et al., **DynamicFL: Federated Learning with Dynamic Communication Resource Allocation**, *IEEE BigData 2024* Best Student Paper
- X. Wang, **Q. Le**, A. F. Khan, et al., **ICL: An Incentivized Collaborative Learning Framework**, *IEEE BigData 2024*
- E. Diao, **Q. Le**, S. Wu, et al., **ColA: Collaborative Adaptation with Gradient Learning**, *arXiv:2404.13844, preprint, 2024*
- **Q. Le**, E. Diao, X. Wang, et al., **Personalized Federated Recommender Systems with Private and Partially Federated AutoEncoders**, *Asilomar 2022*

Honors and Awards

- **Outstanding Paper Award** | ACL 2026
- **Machine Learning Systems Fellowship** | Amazon 2025–2026
- **Best Student Paper Award** | IEEE BigData 2024

Teaching and Professional Service

- **Teaching Assistant**
  University of Minnesota, Twin Cities Mar 2021 – Present
  - Business Analytics; Algorithms and Data Structures; Introduction to Algorithms and Data Structures.
- **Reviewer**: ICLR 2027; NeurIPS 2025; Neural Processing Letters and IEEE BigData 2023; Asilomar 2022.
- **Co-reviewer (assisting faculty)**: USENIX ATC; USENIX FAST; IEEE Transactions on Parallel and Distributed Systems (TPDS).